

Perspectives in Magnetic Resonance

Magnetic Resonance of Porous Media (MRPM): A perspective

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ABSTRACT

Porous media are ubiquitous in our environment and their application is extremely broad. The common connection between these diverse materials is the importance of the microstructure (μm to mm scale) in determining the physical, chemical and biological functions and properties. Magnetic resonance and its imaging modality have been essential for noninvasive characterization of these materials, in the development of catalysts, understanding cement hydration, fluid transport in rocks and soil, geological prospecting, and characterization of tissue properties for medical diagnosis. The past two decades have witnessed significant development of MRPM that couples advances in physics, chemistry and engineering with a broad range of applications. This article will summarize key advances in basic physics and methodology, examine their limitations and envision future R&D directions.

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1. Introduction

Look around your home, offices, laboratories, buildings, and the cities you live in and visit. You will find that our living environment is not only filled with porous materials, but they perform essential functions in our modern society. For example, concrete, the single most used material in the world at two billion tons per year, is an aggregate of complex chemistry and pore structures from nanometer to centimeter scales. Porous stones, wood, composites, and fibrous materials such as papers are a common sight in offices and homes. Near-surface aquifers store 99% of the planet's fresh water in highly porous and unconsolidated rocks. Porous sedimentary rock formations provide much of the primary energy (oil and gas) that powers the modern economy and living standard. The scale of our energy need is one of the most prominent drivers for scientific and technological development to understand and control porous media.

Porous media are generally considered a material with structures of importance at length scales larger than atoms and these structures are essential for the material properties and performance. For example, fluid transport is an essential property for many porous media from water and contamination movement in aquifers, chemical reactions in porous catalysts, filtration, gas exchange in lung tissue, and blood and oxygen perfusion in brain and cancer growth. Magnetic resonance (MR) is uniquely suited for such studies primarily because the pore filling fluid often exhibits a strong MR signal and the power of MR allows direct detection of the fluid through spin relaxation, diffusion, flow and imaging experiments.

MR experiment can be performed without physical contact of the sample. This is an advantage that cannot be overemphasized. Many other analytical techniques require contact, sampling, sample manipulation, single phase or optical transparency. It is particularly powerful when the materials of investigation are large, immobile or optically opaque. Magnetic Resonance of Porous Media (MRPM) has become an exciting branch of science that enjoys broad range of applications and opportunities to engage in multidisciplinary research.

This article will summarize a few aspects of MR research of porous media mainly from a personal perspective. The focus is the essential physical processes that have enabled the use of MR for porous media study and the methodology development in the past two decades. Then I will discuss the limitation of the current methodology and future directions of MRPM. Instrumentation is critical to MRPM and is well covered in a recent book [1]. I will touch upon the instrument need in the last chapter.

1.1. Classical period: 1950–2000

The study of porous materials using MR started from the early days of MR. In 1956, Varian filed a patent describing a method to perform MR at the Earth magnetic field frequency to detect ground water [2]. In the same year, Torrey published a paper entitled "Bloch equations with diffusion terms" [3] laying the foundation to understand the complex diffusion dynamics inside the porous media. Brown [4] considered magnetic field inhomogeneity due to the packing of the grains in a model porous media. From late 1960s and early 1970s NMR relaxation properties have been considered a potential for characterizing permeability of porous rocks [5,6]. The basic physics of the surface relaxation mechanisms has

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also been studied [7,8] and its relationship to pore surface area [9,10]. In 1982, two papers outlined a concept to use spin relaxation to probe surface-to-volume ratio (SVR) [11,12] which is still in wide application today. In late 1970s, Brownstein and Tarr published two classic papers [13,14] combining diffusion and surface relaxation demonstrating the complexity and richness of MR at probing the pore structure. Since then, NMR spin relaxation has found extended applications in the study of many materials, such as cement [15,16], microporous materials such as zeolites [17], and subsurface rock formations [18].

In parallel to the development of the spin relaxation theory, direct measurement of diffusion was also found to be a fruitful approach to characterize pore space. The effect of the pore space on molecular diffusion was recognized in early 1960s [19,20]. Work on restricted diffusion in open pores led to the concept of time-dependent diffusion coefficient [21–24] as the key physics to describe diffusion dynamics in porous media. Callaghan et al. [25,26] discovered the diffusion diffraction phenomenon when the gradient wavelength approaches the characteristic pore size.

For geological prospecting, traditional laboratory MR equipment is clearly inadequate because it requires the sample to be inside the magnet. Inside-out pulsed MR design was pursued in the early 1980s [27,28] using permanent magnet array to produce significant magnetic field outside the instrument. Such idea has grown into a large array of MR tools for well-logging and established an important service for petroleum and mining exploration, water aquifer prospecting [29,30], as well as diverse range of devices for research and concepts [1]. This period was reviewed by Kleinberg in Ref. [31].

1.2. Modern period: 2000–now

One key technology shift in the past decade is the wide adoption of the two-dimensional (2D) and multi-dimensional (MD) MR methodology for relaxation and diffusion. This is similar to the adoption of the 2D and MD methods for Fourier spectroscopy starting in 1970s. In contrast, a 2D-relaxation experiment requires 2D Laplace inversion which is significantly more difficult than its 1D counterpart and MD Fourier inversion. The development of an efficient algorithm for 2D Laplace inversion was the key for the adoption of the 2D methods. Such MD method significantly enhanced the spectral resolution of relaxation and diffusion and has been applied in many research laboratories and commercially in well-logging.

In addition to enhance resolution, 2D methodologies open a new possibility to study the dynamics of spin and molecular motion. For example, chemical exchange and complex diffusion dynamics can be studied by T_2 – T_2 and T_1 – T_2 type of experiments, in a similar fashion as COSY and NOESY.

Biological tissues exhibit complex structures and many aspects are similar to inorganic porous media. For example, trabecular bones are porous and their porous structure is known to be important to the mechanical integrity. The loss of bone and its structure is a key cost of bone fracture especially at advanced ages. Perfusion of tissues of nutrients, oxygen, and drug molecules is important for understanding tissue physiology and drug delivery. Imaging the flow and diffusion of contrast agents using DCE-MRI is in many ways mirroring the experiments to understand fluid flow in subsurface rocks and soils. Imaging of brain microstructure using DTI (diffusion tensor imaging [32]) and DSI (diffusion spectrum imaging [33]) has greatly advanced our understanding of brain structure and neuroanatomy. However, such approaches have not been broadly applied to study inorganic porous media.

2. Physics: surface relaxation and diffusion

2.1. Spin relaxation and diffusion

Spin relaxation is a key parameter describing the dynamics of the spins under investigation and its interaction with the surrounding environment. Many physical processes could contribute to such relaxation. For example, magnetic ions could exhibit a strong fluctuating magnetic field in its immediate surrounding. Such magnetic ion can be a strong relaxation source for the nearby nuclear spins and also those that travel to the close proximity temporarily. In fluid filled porous media, relaxation occurs at the fluid/solid boundary when the fluid spins (such as water molecules) diffuse to the surface. Such behavior can be characterized by a surface relaxivity which is proportional to the ion density. The resulting relaxation rate of the pore fluid is then proportional to SVR which is an important parameter of the pore structure.

In a molecular fluid without magnetic components, the spin relaxation is often dominated by the dipolar interaction between nuclear spins. The magnetic field experienced by one spin from another may vary as the distance and/or the orientation of the pair changes. As a result, both translational and rotational diffusion may introduce a fluctuating magnetic field to cause spin relaxation. In many liquids, dipolar relaxation due to protons on the same molecule is the main source of relaxation. For example, the rotational diffusion of small molecules tends to be fast and the correlation time is short and thus the relaxation is weak. For larger molecules, their rotational correlation time is longer and thus the relaxation is more effective. Hence spin relaxation can be used as a means to determine molecular size.

Molecular translational diffusion shares a similar behavior in bulk fluids with relaxation that smaller molecules diffuse faster and larger molecules slower. Since bulk fluid diffusion constant of many substances is known and well studied, diffusion can be used to identify molecules and fluid phases in a mixture.

2.2. Diffusion and relaxation of chain molecules

Characterization of bulk fluid diffusion is a prerequisite for the understanding of its diffusion in porous media. Diffusion of solid spherical particles can be well described by the Stoke–Einstein relationship. However, for molecules with significant internal structure, such as long chain molecules (e. g. polymers, hydrocarbon chains), the internal molecular motion can have a significant effect on the diffusion properties. For example, the diffusion coefficient is often found to be a power law for polymers and alkanes [34–36].

Generally, the diffusion coefficient of an individual molecule in a mixture can be divided into two parts. One part depends on the individual molecule, such as its size, internal structure and stiffness; and the second part depends on the properties of the surrounding fluid, such as the density and the viscosity. Thus, the diffusion coefficient for the i th component (D_i) should have the form: $D_i = N_i^{-\nu} g(\{N_i\})$, where $g(\{N_i\})$ is a function of all components in the mixture and common for all molecules. Freed et al. [37] have found that this relationship is indeed true for a large number of alkane mixtures that strongly validated the above model. They further showed a complete scaling relationship,

$$D_i = AN_i^{-\nu} \bar{N}^{-\beta}, \quad (1)$$

where $\bar{N}$ is the average chain length. This scaling relationship can be used to obtain mixture composition directly from diffusion and relaxation, for example, using low field NMR [37,38].

2.3. Relaxation in porous media

When a fluid saturates the pore space, interaction between the fluid molecules and the solid surface may contribute to additional relaxation, a phenomenon often called surface relaxation. One source of such relaxation is the presence of magnetic ions on the solid surface. When a water molecule diffuse close the surface, the fluctuating field can cause spin relaxation [7,8,39,40]. Such surface relaxation has been directly measured by drying the pore fluid to only a monolayer on the solid surfaces [15]. Field dependent T_1 of fluid saturated porous media [41–43] show anomalous diffusion dynamics on the solid surface to be characteristic of Lévy walk.

Independent of the mechanism of surface relaxation, the experimentally measured relaxation rate is often found to be proportional to SVR. This may be understood by considering only a surface layer of thickness δ of the fluid is affected by the surface relaxation, to have a relaxation time of T_s . The volume of this layer is δS where S is the total surface area of the sample. Since water molecules will diffuse between this thin layer and the bulk space of the pores (total pore volume of V_p), the measured relaxation will be an average between the surface and bulk volumes,

$$\frac{1}{T_{1,2}} = \frac{1}{T_{1b,2b}} + \frac{\delta S}{V_p} \cdot \frac{1}{T_{1s,2s}} \quad (2)$$

where the subscript 1 and 2 denotes T_1 and T_2 , respectively and the subscript s denotes the relaxation of the surface layer. T_{1b} and T_{2b} are the bulk relaxation of T_1 and T_2 . The above equation is often rewritten in term of surface relaxivity, $\rho \equiv \delta/T_s$,

$$\frac{1}{T_{1,2}} = \frac{1}{T_{1b,2b}} + \rho_{1,2} \frac{S}{V_p} \quad (3)$$

and ρ_1 and ρ_2 denote the surface relaxivity for T_1 and T_2 processes. The validity of the above discussion [9,11] is dependent on a key assumption that the water molecules diffuse rapidly between the surface and the bulk regions so that the relaxation effect is averaged. This is often called the *fast diffusion* assumption. Brownstein and Tarr [14] described an analytical solution of diffusion-relaxation equations in a few simple geometries demonstrating the importance of diffusion in relaxation measurements and interpretation. For arbitrary pore geometries numerical methods can be applied to solve the combined relaxation and diffusion problem with regard to NMR experiments [44,45]. Further theory of diffusion dynamics for NMR will be discussed in Section 4.

2.4. Diffusion in complex media

It is well-known that a porous structure reduces the diffusion of molecules in the pore-filling fluid. This behavior is often characterized by a reduced diffusion coefficient as compared to its bulk value (D_0). Solid porous materials such as rock and soil are, to an approximation, a packing of solid particles with interstitial space occupied by fluids. This is the case in petroleum reservoirs and water aquifers. When water molecules are near a solid pore wall, the space available for the movement of the water molecules is reduced since the solid wall cannot be penetrated by water. For example, if a molecule starts right at the surface, the only allowed movement is away from the wall [46] so that the range of displacement is reduced from that of free diffusion. In biological tissues, cell membranes exhibit finite permeability to water transport thus limiting the diffusion displacement through the membranes and in tissues. Also, some tissues exhibit specific cell shapes, such as the elongated plant cells and extended neurons in animals, the diffusion restriction can be anisotropic. This can give rise to diffusion anisotropy.

For a closed system such as fluid drops, the long time displacement is finite and limited by the size of the enclosing structure. For example, at the limit of long diffusion time when the initial and final positions are uncorrelated, the mean square displacement is

$$\langle \Delta r^2 \rangle \equiv \int \int |\mathbf{r}_1 - \mathbf{r}_2|^2 P(\mathbf{r}_1, \mathbf{r}_2, t) d\mathbf{r}_1 d\mathbf{r}_2 \quad (4)$$

$$= \frac{1}{V_p} \int \int |\mathbf{r}_1 - \mathbf{r}_2|^2 d\mathbf{r}_1 d\mathbf{r}_2 \quad (5)$$

$$= 2(\langle r^2 \rangle - \langle r \rangle^2), \quad (6)$$

where P is the diffusion propagator and $P = 1/V_p$ at long time (V_p is the pore volume). For a sphere of diameter d , $\langle \Delta r^2 \rangle$ can be evaluated and is given by

$$\langle \Delta r^2 \rangle = 3d^2/5. \quad (7)$$

For NMR experiments measuring diffusion along one direction, the mean-square displacement will be one-third of $\langle \Delta r^2 \rangle$. This phenomenon has been used to characterize emulsions [47,48].

For an open porous system where the pore space is interconnected, there are two aspects of this effect. One is that at short diffusion time, those molecules near the surface of the solid matrix (or grains) have less free space to diffuse and the net displacement is reduced. This effect is directly proportional to SVR of the material and it is often characterized by a reduced diffusion coefficient [23],

$$\frac{D(t)}{D_0} \approx 1 - A\sqrt{D_0 t} \frac{S}{V_p}, \quad (8)$$

where D_0 is the bulk diffusion coefficient, and $D(t)$ is the diffusion coefficient at the diffusion time t . A is a constant dependent on dimensionality. The approximation is valid when the diffusion distance $\sqrt{D_0 t}$ is much less than pore size. Here the diffusion coefficient is defined as $\langle \Delta x(t)^2 \rangle / 2t$, where $x(t)$ is the net displacement along one direction at time t and $\langle \Delta x(t)^2 \rangle$ is the mean square displacement. For isotropic diffusion in three dimensions, $D(t)$ can be defined from the vector displacement $\Delta \mathbf{r}$: $D(t) = \langle \Delta \mathbf{r}(t)^2 \rangle / 6t$.

The second effect is the long time diffusion when the diffusion distance, $\sqrt{D_0 t}$ is larger than the characteristic length scale of the pores, such as the pore size. For an open system, $D(t)$ will approach a constant at long time, but with a value less than D_0 . This is because the diffusing molecules have to travel around the grains so that the distance to travel in order to achieve a displacement x is larger than x . In other words, such molecules have to meander around the grains (geometrical obstacles) in a similar way as motorists have to travel along predetermined streets instead of moving straight to destinations. The reduction of the long time diffusion coefficient compared to D_0 is related to the tortuosity factor T [21],

$$\frac{D(t \rightarrow \infty)}{D_0} = \frac{1}{T}. \quad (9)$$

For more theoretical review of the physics of diffusion in porous media, please refer to the papers by Sen [49] and Grebenkov [50].

2.5. Diffusion in internal field

The diffusion coefficient is often measured by applying known magnetic field gradients. When a porous sample is placed in a uniform magnetic field (such as the NMR/MRI magnet), the magnetic field inside the sample, in particular, inside the pore space, is naturally inhomogeneous due to the susceptibility contrast between the solid phase and the fluid in the pores. Molecular diffusion in such an internal field can be detected by NMR and it has been shown to reflect the underlying pore geometry [51–54]. The spatial profile of this field will be determined by the structure of the underlying

porous medium. While this assertion is qualitatively not surprising, in fact, quantitative connections exist between the pore structure and the magnetic field structure. For example, the two-point correlation function of the internal field is closely related to the structural factor of the medium in a model system [55,56]. Experimental method has been developed to specifically use this internal field to extract pore size distributions in inorganic porous materials and tissues [57,58].

2.6. Diffusion anisotropy

If the enclosing structure exhibits anisotropy, such as a long cylinder, the diffusive motion of the molecules will reflect this confinement symmetry. Considering the long cylinder with a diameter r and a length L , and $L \gg r$. When the diffusion time Δ is such that $\sqrt{D\Delta} \sim r$, the diffusion across the cylinder is significantly restricted leading to an apparent diffusion constant much reduced from its bulk value. However, along the axial direction, $\sqrt{D\Delta} \ll L$, thus the axial extent of the structure is not reflected in the axial diffusion and the axial diffusion constant will remain close to the bulk diffusion coefficient. This diffusion anisotropy can be approximated by a second rank tensor. In biomedical imaging, this diffusion anisotropy is often used to characterize the fiber structure of tissues through the technique of diffusion tensor imaging (DTI) [59,32].

3. Multidimensional NMR of relaxation and diffusion

Conventional NMR relaxation and diffusion techniques measure one parameter (e.g. T_1 and T_2 , $T_{1\rho}$, or D) in a single experiment. For example, T_2 is typically obtained by executing a series of NMR experiments with changing relaxation time period, t , and measuring the signal at each t . For a single species of spins, the signal decays in an exponential functional form. For mixtures, each component may have a unique T_2 value and thus the total signal (M) will be a superposition of many exponential functions,

$$M(t) = \sum_i m_i \exp(-t/T_{2i}) \quad (10)$$

where T_{2i} is the relaxation time constant and m_i the amplitude of the signal from component i . For example, in a fluid mixture of different molecules, the smaller molecules often exhibit faster translational and rotational diffusion and thus longer T_2 . Thus, the distribution of T_2 can be used to quantify molecular weight distributions. In porous media, T_2 is found to be enhanced by surface relaxation and thus used to measure pore sizes. However, in a more complex sample, such as a porous rock saturated with a mixture of water and oil, the relaxation time constant of water and oil may be similar due to a combination of oil molecular size and water surface relaxation.

The newly developed multi-dimensional (MD) NMR technique improves the resolution of different molecules by measuring the correlation function of two or more parameters, such as T_1 – T_2 correlation, D – T_2 , T_1 – T_2 – D , and D – D . The one-dimensional (1D) NMR experiment typically refers to an experiment that measures signals as a function of a particular experimentally controlled parameter, such as a time delay in the pulse sequence, the amplitude of an RF pulse, or the amplitude of the field gradient pulse. In contrast, MD NMR is performed by varying two or more experimental parameters and acquires NMR signals as a two- or higher dimensional data matrices. Laplace inversion of such MD matrices is used to obtain the multi-parameter correlation functions. Such MD correlation spectrum allows more separation of signals from different components, helps to identify different phases and to observe changes of the composition and physical properties.

For example, a T_1 – T_2 correlation experiment uses two parameters: t_a for the T_1 recovery time, and t_b for the T_2 decay time. Typically, for every value of t_a , data at the full range of t_b is acquired. In this case, the final data set is a full 2D matrix. Other experiments may not require to fill the entire 2D data matrix.

This type of correlation experiments was first reported in 1980s to study the multiple components of a chemical and biological systems. Even though the NMR experiments of this type could be performed on the NMR spectrometer at the time, data analysis was a considerable barrier due to the requirement of large computing power and computer memory [60]. In early 2000, a new algorithm was developed to rapidly perform the multi-dimensional Laplace inversion without the huge memory requirement so that MD NMR data can be analyzed on regular PCs and laptops [61]. This development leads to a rapid expansion of application of the MD methodology in geological materials [61–63], chemistry [64], material science [65,66], food science [67,68]. It has been since used commercially in petroleum exploration [69].

3.1. Relaxation correlation experiments: T_1 – T_2

The pulse sequence for a T_1 – T_2 experiment is the following:

$$RD - \underbrace{\pi - t_a}_{\text{first part}} - \underbrace{\frac{\pi}{2} - t_e/2 - \left[\pi - \frac{t_e}{2} - \text{Acq} - \frac{t_e}{2} \right]_N}_{\text{second part}} \quad (11)$$

where N is the echo number, RD the recycle delay, and Acq is data acquisition. The first time variable is t_a and the second one is $t_b = Nt_e$. This sequence is essentially a combination of the inversion-recovery and the CPMG, thus it is often called IRCPMG. The CPMG is used to detect the signal of T_2 decay from the end of the T_1 recovery period. For a spin species with a specific T_1 and T_2 , the signal is proportional to $[1 - 2\exp(-t_a/T_1)]\exp(-t_b/T_2)$, thus the total signal as a function of t_a and t_b is

$$M(t_a, t_b) = \iint (1 - 2e^{-t_a/T_1})e^{-t_b/T_2} \times \mathcal{F}(T_1, T_2) dT_1 dT_2, \quad (12)$$

where $\mathcal{F}(T_1, T_2)$ is the probability density of molecules with specific T_1 and T_2 . To obtain $\mathcal{F}$ from data M , a 2D Laplace inversion is needed.

3.2. Relaxation correlation experiments: T_2 – T_2

The general concept of a 2D NMR experiment is flexible to include arbitrary spin evolution for the first and second periods. The T_2 – T_2 experiment is an interesting example that both periods involve T_2 relaxation process. The pulse sequence is shown below

$$\underbrace{\frac{\pi}{2} - \left[\frac{t_e}{2} - \pi - \frac{t_e}{2} \right]_n}_{T_2 \text{ relaxation } (T_2^a)} - \underbrace{\frac{\pi}{2} - \Delta - \frac{\pi}{2} - \frac{t_e}{2} - \left[\pi - \frac{t_e}{2} - \text{Acq} - \frac{t_e}{2} \right]_m}_{T_2 \text{ relaxation } (T_2^b)} \quad (13)$$

The first segment involves T_2 relaxation through the use of CPMG sequence with n π pulses and the echo signals is not measured. The second $\pi/2$ pulse at the end of the $[\dots]_n$ rotates the magnetization back to the z -axis. It is important to note that this magnetization has gone through T_2 relaxation for this period. During the subsequent delay time of Δ , the spins experience T_1 relaxation since they are aligned with $\mathbf{B}_0$. The second period is a CPMG sequence of m π pulses and the echo train is acquired. In this experiment, the two T_2 relaxation evolutions are measured by two independent times, $t_a = nt_e$ and $t_b = mt_e$.

The inclusion of the time Δ is important for this experiment because the two T_2 variables are measured at different times (T_2^a and T_2^b). If the system is static, that is the T_2 of a molecule is identical before and after Δ , then $T_2^a = T_2^b$. On the two-dimensional plot of

T_2^a vs. T_2^b , only diagonal signals (i.e. $T_2^a = T_2^b$) will appear and there will be no off-diagonal signals ($T_2^a \neq T_2^b$). If, however, during Δ , the T_2 of a molecule is changed, then an off-diagonal signal appears. Such an experiment thus has the potential to characterize the complex system dynamics (e.g. Refs. [64,70]). Since T_1 relaxation is effective during Δ period, the range of this period is limited by T_1 .

Similar concepts have been demonstrated recently for other relaxation processes, such as T_1 - T_1 experiment, $T_{1\rho}$ - $T_{1\rho}$ experiment. These different experiments examine different relaxation mechanisms that could be sensitive to other aspects of molecular dynamics. They are an interesting extension of the T_2 - T_2 concept and their potentials are to be explored.

3.3. Diffusion-relaxation correlation experiments

Fig. 1 illustrates two implementations of the diffusion-relaxation experiment using pulse field gradients. In the first implementation, a spin-echo sequence is used for segment 1 with a gradient pair before and after the second pulse. The first echo spacing t_{cp1} is often quite long, several milliseconds, in order to allow sufficient time for the gradient pulses to yield an appropriate diffusion decay given the diffusion constant of the sample. The second echo spacing during segment 2 is often very short, for example, 100 μ s, to increase the number of echoes and to reduce the effect of background gradients. The diffusion decay caused by this segment is known [71] as

$$\exp[-D\gamma^2 g^2 \delta^2 (\Delta + 2\delta/3)], \quad (14)$$

where γ is the gyromagnetic ratio, D the diffusion constant, g the amplitude of the gradients, δ the duration of the gradient pulse, and Δ the time period between the two gradients. This equation is valid within the Gaussian phase approximation, such as diffusion in bulk fluids.

A stimulated echo with pulsed gradients can also be used for the diffusion weighting segment, Fig. 1B. The first three 90° pulses and the two associated gradient pulses produce the stimulated echo. The 180° pulses afterward refocus the magnetization repeatedly producing a train of echoes. Similar to the spin-echo case for the diffusion weighting (Fig. 1A), the period between the first two pulses is often several milliseconds long to allow sufficient gradient-induced modulation. In addition, the diffusion time between the 2nd and 3rd pulses ($\sim \Delta$) allows a further increase of diffusion

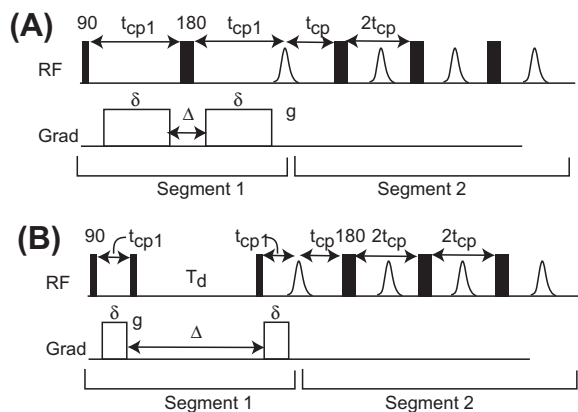


Fig. 1. Diffusion-relaxation correlation sequences using pulsed field gradients. (A) The first segment is a spin-echo with the echo appearing at a time $2t_{cp1}$ after the first pulse. (B) The first segment is a stimulated echo appearing at a time t_{cp1} after the third pulse. The detection (2nd) segment for both is a CPMG pulse train. The amplitude or the duration of the gradient pairs in both sequences is incremented in order to vary the diffusion weighting.

weighting. A long Δ period can be particularly useful for samples with a short T_2 but a long T_1 .

For the case of the spin-echo for diffusion weighting, the full kernel, k , can be written as

$$k = \exp(-2t_{cp1}/T_2) \times \exp[-D\gamma^2 g^2 \delta^2 (\Delta + 2\delta/3)] \times \exp(-2nt_{cp}/T_2), \quad (15)$$

where the first factor is the T_2 decay during the first segment which can be a fixed constant provided t_{cp1} is kept constant for different g 's. The full kernel for the stimulated echo case can be written as

$$k = \exp(-2t_{cp1}/T_2 - T_d/T_1) \times \exp[-D\gamma^2 g^2 \delta^2 (\Delta + 2\delta/3)] \times \exp(-2nt_{cp}/T_2). \quad (16)$$

The first factor is the T_1 and T_2 decay during the first segment. In both cases, the kernel can be written as a tensor product.

A static field gradient can also be used to measure diffusion [72,73]. A static field gradient is always present in regions away from the geometric center of most NMR magnets, and also in specifically designed magnets using permanent magnets, such as the NMR well-logging tool [74], Garfield magnet [75] and the NMR MOUSE [76]. Such permanent magnet can be miniaturized, inexpensive, and there is no need for pulsed field gradients, significantly simplifying the equipment and operation of diffusion experiments.

3.4. Diffusion-diffusion correlation experiment

Similar to the relaxation-relaxation correlation technique, diffusion can be used for the both dimensions of the 2D experiment. Using pulsed field gradients, the pulse sequence is shown in Fig. 2. Such experiment can also be performed with spin echo sequence, for example, using the segment 1 in Fig. 1A for both segment. The one advantage of the stimulated echo based sequence (Fig. 2) is its flexibility in adjusting T_d . For either case, we use b_1 and b_2 to denote the diffusion weighting for segment 1 and 2, the kernel can be written as

$$k = \exp[-D_1 b_1 - D_2 b_2] \quad (17)$$

Here we have ignored the relaxation contribution to the kernel since they are kept constant in an experiment.

Both segments measure the diffusion coefficient, however, at different times separated by T_d . We will use D_1 and D_2 to denote the diffusion coefficient associated with the first and second segments (gradient g_1 and g_2), respectively. If a molecule does not change its diffusion coefficient during T_d , then $D_1 = D_2$. If the molecule changes its own configuration of environment during T_d , then $D_1 \neq D_2$. On a 2D plot with axes D_1 and D_2 , signals with $D_1 = D_2$ are on the diagonal line and the that with $D_1 \neq D_2$ are off-diagonal. As a result, the off-diagonal signal can be an indication of molecular exchange and dynamics.

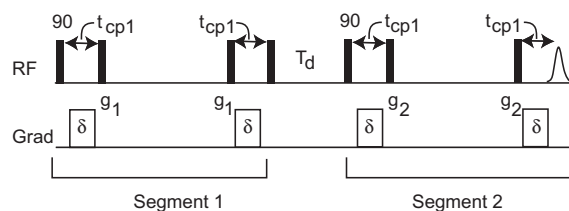


Fig. 2. Diffusion-Diffusion correlation sequences using pulsed field gradients. The two segments are both a stimulated echo sequence. At the end of the first stimulated echo, a $\pi/2$ pulse is used at a time t_{cp1} after the third pulse in order to rotate the stimulated echo magnetization to the longitudinal direction. The 2nd segment detects the stimulated echo. All RF pulses are $\pi/2$. The amplitude or the duration of the two gradient pairs (g_1 and g_2) are incremented independently.

To include this exchange effect, the signal of the experiment can be written as

$$M(b_1, b_2) = \int \int \exp[-D_1 b_1 - D_2 b_2] \times \mathcal{F}(D_1, D_2; T_d) dD_1 dD_2, \quad (18)$$

where $\mathcal{F}(D_1, D_2; T_d)$ is the probability density of molecules that evolve from D_1 to D_2 over the time period T_d , a concept similar to that of the diffusion propagator. In an homogeneous system, the correlation vanishes at long time so that $\mathcal{F}(D_1, D_2, T_d \rightarrow \infty) = F(D_1)F(D_2)$, where $F(D)$ is the 1D diffusion distribution. Thus the 2D plot of D_1 and D_2 will be very broad showing extended exchanges between all values of diffusion.

On the other hand when T_d is much shorter than the characteristic exchange time, then D_1 and D_2 are highly correlated, thus only allowed signal will satisfy $D_1 = D_2$, and only diagonal signal can be observed. Hence, the off-diagonal amplitude and its T_d dependence can be used as a measure of the exchange phenomenon. In fact, this behavior is very similar to the conventional NOESY experiment in 2D Fourier spectroscopy.

3.5. Exchange or diffusion

It is important to distinguish this type of chemical exchange, such as a chemical reaction, or a molecular configuration modification, from diffusion dynamics. Diffusion in porous media, in particular, diffusion between different pores, is often discussed in the literature loosely as exchange, since it is often described as the fluid molecules “exchange” between different pores. However, the behavior of chemical exchange and diffusion is fundamentally different in a 2D NMR experiment. The mechanism of chemical exchange is external to the spin Hamiltonian and its spin evolution so that it can be introduced as an independent probability ($\mathcal{F}(D_1, D_2; T_d)$). However, for a stationary diffusion system, such as fluid filled pore space, the spin Hamiltonian considers diffusion dynamics [44] fully and explicitly through Torrey equation, Eq. (19), for all relaxation and diffusion pulse sequences. Thus, diffusion between different pores should not be explicitly considered as an independent probability. Also, the diffusion theory predicts several features that cannot be explained by the exchange formalism. The theory of 2D NMR for diffusion system is discussed in Section 4.

4. Theory of 2D NMR of diffusion systems

4.1. Diffusion equation and eigenmode methods

The magnetization evolution due to diffusion and relaxation can be studied theoretically using the Bloch-Torrey equation [3]:

$$\frac{\partial}{\partial t} m(r, t) = D \nabla^2 m(r, t) - \mu m(r, t). \quad (19)$$

Here $m(r, t)$ is the relative difference of the magnetization and its equilibrium value, D is the bulk diffusion coefficient, and μ is the bulk relaxation rate. Also, the spin precession is not considered here because it is absent during T_1 decay and being refocused during CPMG. The general solution to the Torrey-Bloch equation can be written as a sum of eigenmodes,

$$m(r, t) = e^{-\mu t} \sum_{i=0}^{\infty} a_i \phi_i(r) e^{-t/\tau_i}, \quad (20)$$

where ϕ_i and τ_i are normalized eigenfunctions and eigenvalues, respectively. ϕ_i with different indices are orthogonal to each other, and satisfy the following equation:

$$D \nabla^2 \phi_i(r) + \frac{1}{\tau_i} \phi_i(r) = 0. \quad (21)$$

The pore boundary behaves as a magnetization sink due to the surface relaxation thus the diffusion flux into the surface is related to the strength of the surface relaxivity ρ :

$$D \hat{n} \cdot \nabla \phi_i + \rho \phi_i = 0, \quad (22)$$

where $\hat{n}$ is the unit vector normal to the surface. Once the location of the boundaries are known, the above equation can be applied to determine the eigenvalues. Analytical and numerical solutions and the properties of these solutions have been reported in the literature [14,77,78,45,79].

Since the T_1 and T_2 processes may have contributions from different mechanisms, thus the surface relaxivity is in general different for T_1 and T_2 , so we label them as ρ_1 and ρ_2 , respectively. As a result, we need to consider two sets of eigenmodes corresponding to the T_1 and T_2 dynamics. We call the two sets of modes L -modes and T -modes and denote them ϕ_i^L and ϕ_i^T and their eigenvalues T_{1i} and T_{2i} , respectively [44]. For example, during a inversion recovery experiment, ρ_1 and ϕ_i^L are modes to describe the spin dynamics, while for CPMG, ρ_2 and ϕ_i^T .

Here we show the solution for one-dimensional model to show an intuitive understanding of the eigenmodes. In a one-dimensional pore, the Torrey-Bloch equation can be solved analytically and with fast diffusion and weak relaxation, it can be shown that the eigenvalues are [14]

$$\tau_0 \approx \frac{d}{\rho} \quad (23)$$

$$\tau_i \approx \frac{d^2}{D\pi^2 i^2}, i = 1, 2, 3, \dots \quad (24)$$

where d is the pore size, ρ the surface relaxivity. The eigenfunctions $\phi_i(r)$ for this case are readily calculated. The “ground state” eigenfunction $\phi_0(r)$ is approximately constant across the pore, while the higher order functions ($i > 0$) oscillate in space [13,80,79].

Any magnetization, $m(r, t)$, can be decomposed into a sum of the eigenmodes, Eq. (20). The weight of these “modes” in the magnetization is given by a coefficient a_i , which is determined by the overlap integral of $m(r, t)$ with the i -th eigenmode, $a_i = \int dr m(r, 0) \phi_i(r)$. This weighting can be selectively manipulated to some extent through the choice of pulse sequence. Since the B_1 field is not spatially varying over pore scale as in a typical NMR probe, the RF pulses do not usually excite higher modes directly. It is typically the spin evolution during the relaxation time or precession that creates the spatial modulation and thus higher modes. This scenario is similar to the excitation of multiple quantum coherences in conventional Fourier NMR spectroscopy that relies on the non-commuting Hamiltonian such as dipolar coupling. The more control is exercised on which modes are “excited”, the less error will be associated with the data analysis for extracting pore microstructural properties. For example in fast diffusion limit ($\rho d/D \ll 1$), an inversion recovery experiment will only excite the lowest mode, $a_0 \neq 0$ [14]. This is because in fast diffusion, the lowest mode is approximately a constant in the pore space, $\phi_0 \sim 1/V_p$. Such a flat ϕ_0 matches well with the spatially flat magnetization profile ($m(r, 0)$ at time zero) on the pore length scale. Thus $a_0 \approx 1$. Because the higher modes are orthogonal to ϕ_0 , $a_i = \int dr m(r, 0) \phi_i(r) \approx \int dr \phi_0 \phi_i = 0$ when $i \neq 0$. The ground state mode is sometimes referred to as the relaxation mode for this reason. The decay of the modes of $i > 0$ is primarily due to diffusion, hence the name “diffusion” modes.

It should be noted that the decomposition shown in Eq. (20) is not necessarily a subdivision of spins, since all spins in general are subject to both relaxation and diffusion. Rather, it is a classification of different spatial structures of magnetization modulation and their corresponding decay dynamics.

4.2. 2D NMR of diffusion system

We will use the eigenmode formalism to describe the diffusion dynamics during a 2D NMR experiment including the pulsed field gradient effect. The following examples will illustrate the application of the eigenmode formalism to analyze pulse sequences. We consider that the RF magnetic field is uniform over the pore length scale so that the RF pulses do not introduce additional spatial modulation. Also, all eigenmodes will be considered and the formalism is applicable for all diffusion regimes, not limited to fast diffusion.

4.2.1. Relaxation correlation

A T_1 – T_2 experiment using IRCPMG as shown in Eq. (11) involves two periods with different spin dynamics. During the t_a period, the magnetization dynamics (T_1 process) is described by L -modes. The magnetization, $m_1(\mathbf{r}, t_a)$, at the end of the t_a period is

$$m_1(\mathbf{r}, t_a) = \sum_{i=0}^{\infty} a_i \phi_i^L(\mathbf{r}) e^{-t_a/T_{1i}}. \quad (25)$$

Here a_i is determined by the initial magnetization after the first 180° pulse, $m_0(\mathbf{r}) = -1$. Since different eigenmodes evolves at different rate, the magnetization $m_1(\mathbf{r}, t)$ may develop significant spatial variation as t evolves. After the 90° pulse, the magnetization is rotated to the transverse plane so that the spin dynamics is governed by T -modes. Thus, the evolution of the CPMG echo signal, $m(\mathbf{r}, t_a, t_b)$, is:

$$m(\mathbf{r}, t_a, t_b) = \sum_{j=0}^{\infty} b_j \phi_j^T(\mathbf{r}) e^{-t_b/T_{2j}}. \quad (26)$$

where $b_j = \int m_1(\mathbf{r}, t_a) \phi_j^T(\mathbf{r}) d\mathbf{v}$. The detected time-domain signal is simply the integral of the magnetization:

$$S(t_a, t_b) = \int d\mathbf{v} \cdot m(\mathbf{r}, t_a, t_b) \quad (27)$$

$$= \sum_{ij} s_{ij} e^{-t_a/T_{1i} - t_b/T_{2j}} \quad (28)$$

where $s_{ij} = \langle \phi_j^T | 1 \rangle \langle \phi_j^T | \phi_i^L \rangle \langle \phi_i^L | 1 \rangle$. Here we use the notation $\langle \phi_j | \psi_i \rangle = \int d\mathbf{v} \cdot \phi_j^* \psi_i$, and ϕ_j^* is the complex conjugate of ϕ_j . Also, $|1\rangle$ stands for a constant function within the pore space. The 2D spectrum can be obtained after Laplace inversion as

$$s_{12}(T_1, T_2) = \sum_{ij} \langle 1 | \phi_j^T \rangle \langle \phi_j^T | \phi_i^L \rangle \langle \phi_i^L | 1 \rangle \times \delta(T_1 - T_{1i}) \delta(T_2 - T_{2j}). \quad (29)$$

The signal corresponding to $i = j$ are called diagonal, and those for $i \neq j$ are off-diagonal signals. One of the important feature of the off-diagonal signals is that it measures the overlap integral of different eigenmodes (T - and L -modes). For example, if $\rho_1 = \rho_2$, then the two sets of T - and L -modes are identical, thus there will be no off-diagonal peaks. This is because of the orthogonality of eigenmodes, and $\langle \phi_j | \phi_i \rangle = 0$ (for $i \neq j$) [44].

The same framework can be used to analyze T_2 – T_2 correlation experiment Eq. (13) with the resulting 2D spectrum [44]:

$$s_{22}(T_2^a, \Delta, T_2^b) = \sum_{p,q,r} \langle 1 | \phi_p^T \rangle \langle \phi_p^T | \phi_r^L \rangle \langle \phi_r^L | \phi_q^T \rangle \langle \phi_q^T | 1 \rangle e^{-\Delta/T_{1r}} \times \delta(T_2^a - T_{2p}) \delta(T_2^b - T_{2q}). \quad (30)$$

In contrast to the s_{12} , s_{22} now has a user controlled parameter Δ which can be used to manipulate the weight distribution among the modes. When $\Delta = 0$, since $\sum_r |\phi_r^L\rangle \langle \phi_r^L| = 1$,

$$s_{22}(T_2^a, 0, T_2^b) = \sum_{p,q} \delta_{p,q} \langle 1 | \phi_p^T \rangle \langle \phi_p^T | 1 \rangle \times \delta(T_2^a - T_{2p}) \delta(T_2^b - T_{2q}), \quad (31)$$

where only the diagonal signals survive. When Δ is long, the slowest T_1 mode ($r = 0$) dominates the evolution and the weight distribution will be controlled by the non-orthogonality $\langle \phi_p | \phi_0^L \rangle$ between the T_1 (L) and T_2 (T) modes.

For both T_1 – T_2 and T_2 – T_2 experiments, the presence of the off-diagonal peak requires $\rho_1 \neq \rho_2$, independent of the diffusion regime. This result is valid for any pore systems independent of pore size and connectivity [44,81], a clear distinction from the phenomenon of molecular exchange. Even though the 1D pore is a very simple model, it nevertheless illustrates many of the key features. Ref. [44] has reported calculations of the 2D relaxation spectrum for a range of surface relaxivity and timings to demonstrate the behavior of off-diagonal signals. For example, it shows that in both extreme fast or slow diffusion regimes, since the shape of the eigenmodes becomes insensitive to ρ , then the two sets of T - and L -modes are essentially identical and thus the amplitude of the off-diagonal peaks approaches zero. For a simple 1D pore, a complex off-diagonal signal with several peaks (some positive and others negative) may exist when $\rho d/D \sim 1$ and $\rho_1 \neq \rho_2$.

Another interesting case of coupled pores was studied [44,80] showing that diffusion between pores can be detected by the 2D NMR methods. Fig. 3 shows T_1 – T_2 spectra of a two-pores system and the diffusion coupling is characterized by an effective distance between them. At strong coupling, the two pores essentially form a single pore and the 2D spectrum is a single peak. For very weak coupling, right panel, the two pores are essentially unrelated and thus the 2D spectrum is simply the superposition of the two pores. In the intermediate coupling case, the diffusion dynamics are more complex and the spectrum contains more pronounced off-diagonal peaks. In this case, an observation of the off-diagonal signal should be regarded as indicative of direct diffusion between different parts of the pore space.

4.2.2. Diffusion experiments

A PFG-CPMG experiment such as shown in Fig. 1 involves two periods with different spin dynamics. During the first period, the magnetization is first modulated by the gradient pulse, then demodulated by the same gradient after a time period Δ . The

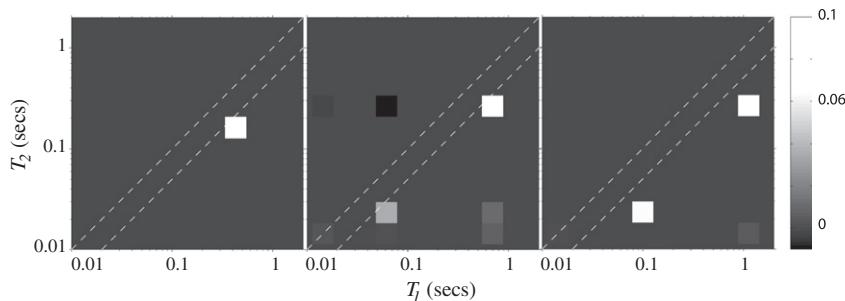


Fig. 3. 2D NMR T_1 – T_2 spectra of coupled pores for different coupling lengths: strong (left), intermediate (middle), and weak coupling (right). Figure from [44].

magnetization, $m_1(\mathbf{r}, \tau_1)$, at the end of the second gradient pulse period is

$$m_1(\mathbf{r}, \Delta, \mathbf{q}) = \sum_{j=0}^{\infty} a_j \phi_j(\mathbf{r}) e^{-i\mathbf{q}\mathbf{r}} e^{-\Delta/T_{kj}}, \quad (32)$$

where $a_j = \langle 1 | e^{-i\mathbf{q}\mathbf{r}} | \phi_j \rangle \equiv \int e^{-i\mathbf{q}\mathbf{r}} \phi_j(\mathbf{r}) d\mathbf{r}$. Depending on the PFG pulse sequences, such as spin echo or stimulated echo, the corresponding eigenmode and decay time constant T_{kj} will be used (ϕ^T and T_{2j} for spin echo and ϕ^L and T_{1j} for stimulated echo).

The second period is a CPMG sequence so that the spin dynamics is governed by ϕ^T modes. Thus, the evolution of the echo signal, $m(\mathbf{r}, \Delta, \mathbf{q}, t)$, is:

$$m(\mathbf{r}, \Delta, t) = \sum_{j=0}^{\infty} b_j \phi_j^T(\mathbf{r}) e^{-t/T_{2j}}. \quad (33)$$

where b_j is determined by $m_1(\mathbf{r}, \Delta, \mathbf{q})$:

$$b_j = \int m_1(\mathbf{r}, \tau_1, \mathbf{q}) \phi_j^T(\mathbf{r}) d\mathbf{v}. \quad (34)$$

The detected time-domain signal is the integral of the magnetization over the sample volume:

$$S(\Delta, \mathbf{q}, t) = \int d\mathbf{v} \cdot m(\mathbf{r}, \Delta, \mathbf{q}, t) = \sum_{ij} s_{ij}(\mathbf{q}) e^{-\Delta/T_{ki} - t/T_{2j}} \quad (35)$$

where $s_{ij}(\mathbf{q}) = \langle 1 | e^{i\mathbf{q}\mathbf{r}} | \phi_i \rangle \langle \phi_j | e^{-i\mathbf{q}\mathbf{r}} | \phi_j^T \rangle \langle \phi_j^T | 1 \rangle$.

5. Laplace inversion

Spin relaxation and diffusion are typically manifested as decaying signals. Data analysis then often involves a Laplace inversion to obtain a spectrum (or distribution) of relaxation times or diffusion constants. This inversion is ill-conditioned in the sense that small noise in the data can cause significant changes in the resulting spectrum. For experimental data with noise, the ill-conditioned nature also manifests itself as the existence of infinite number of solutions that satisfy the noise model. For such ill-conditioned inversion problems, it is most desirable to quantify the properties of the entire solution space [82]. However, the conventional approaches aim at finding a single best solution according to a certain criterion, such as stability against noise.

One common class of methods is designed to select a subset of the solutions, for example, the smooth solutions, and then look for the best solution within this subset. Commonly used methods include Tikhonov regularization [83] and the maximum entropy method [84] to handle this and other types of inversion [85]. Even though there are well-established algorithms to perform numerical Laplace inversion for NMR work [86–88], however, its use may not be straightforward and requires experience. For example, the choice of the optimal regularization parameter is nontrivial. This is primarily due to the complex noise propagation in such an inversion. It is thus useful to understand the essential mathematics involved to better guide its application and be aware of its limitations.

5.1. General theory

The 1D Laplace inversion is to solve the following problem: given a dataset $M(\tau)$ of finite number of data points, determine $\mathcal{F}(T)$, so that $M(\tau) = \int e^{-\tau/T} \mathcal{F}(T) dT$, within the statistics of the noise. Here T is the decay time constant. This equation can be approximated by a discretized matrix form:

$$M = KF + N, \quad (36)$$

where M and N are the data and noise vectors, matrices K , and F are discretized version of the kernel, and $\mathcal{F}(T)$, respectively. Only M and

K are known. The true solution F should satisfy Eq. (36) by $\|M - KF\| < \sigma$, where $\|\cdot\|$ is the vector norm and σ the noise variance. However, given a finite SNR, many solutions satisfy this criterion, and this is the source of the ill-conditioned nature of the Laplace inversion.

The regularization method obtains a fit to the data through minimization of the following expression:

$$\|M - KF\|^2 + \alpha \|F\|^2, \quad (37)$$

The first term measures the difference between the data and the fit, KF . The second term is a Tikhonov regularization and its amplitude is controlled by the parameter α . The effect of this regularization term is to select out a solution with a small 2-norm $\|F\|^2$ and as a result a solution that is smooth and with fewer sharp spikes. However, it may cause a bias to the result. When α is chosen such that the two terms are comparable, the bias is considered minimized and the result is stable in the presence of noise [88,61]. When α is much smaller, the resulting spectrum F can be spiky and sensitive to noise.

In order to accelerate the minimization of Eq. (37), the data and the kernel can be compressed to a fewer number of variables using singular value decomposition (SVD) of K ,

$$K = U \Sigma V^T, \quad (38)$$

where Σ is a diagonal matrix with singular values in a descending order, U and V are unitary matrices. V^T denotes the transpose of matrix V . For the exponential kernel, the singular values typically decay quickly. We limit our algorithm to the subspace spanned by typically fewer than 10 largest singular values. Such a subspace is adequate for the limited signal-to-noise (SNR) ratio of the usual experimental data. Using the SVD of K , Eq. (37) can be rewritten in an identical structure but with a compressed data $\tilde{M} = U^T M U$ and a kernel of much smaller size. Then, the optimization of Eq. (37) is performed in this reduced space resulting in a greatly increased inversion speed.

5.2. Fast Laplace Inversion – FLI

In fact, the 2D Laplace inversion, such as Eq. (12), can be cast into the 1D form of Eq. (36). However, the size of the kernel matrix will be huge. For example, a T_1 – T_2 experiment may acquire 30 T_1 –recovery points and 8192 echoes for each recovery. Thus, assume that 100 points for T_1 and T_2 are used, respectively, the kernel will be a matrix of $(30 \times 8192) \times 10,000$ with 2.5×10^9 elements. It is difficult to employ the commonly used SVD routine (such as that of Matlab) to perform SVD on such a matrix on current desktop computers. Thus the 1D algorithm cannot be used directly in practice.

Many 2D NMR experiments such as most of those outlined earlier exhibit a unique kernel structure such that the full kernel can be written as a product of two independent kernels:

$$k(x_1, x_2, R_1, R_2) = k_1(x_1, R_1) k_2(x_2, R_2). \quad (39)$$

Here x denotes experimental pulse sequence parameters (such as delay time t), and R the spectral parameters (such as T_1). For example, the kernels for the T_1 – T_2 experiment are

$$k_1 = 1 - 2 \exp(-t_a/T_1) \quad (40)$$

$$k_2 = \exp(-t_b/T_2). \quad (41)$$

Thus, if experiments are designed in such a way that the data can be written in the form of a matrix M , Eq. (36) can be then put into the following form:

$$M = K_1 F K_2^T + N. \quad (42)$$

Here M and N are the data and the noise in their 2D matrix form. The matrix elements are defined as $(K_1)_{ij} = k_1(x_{1i}, R_{1j})$ and $(K_2)_{ij} = k_2(x_{2i}, R_{2j})$.

The benefit of this tensor product structure of the kernel is that the SVD of K_1 and K_2 is quite manageable on current computers. Once the SVD of K_1 and K_2 is obtained, the SVD of the product matrix can be approximated. For example, the product of the singular values of K_1 and K_2 will be the singular value of the product matrix K . As a result, for a 2D experiment with similar SNR as a 1D experiment, for example, $\text{SNR} \sim 1000$, then 50–100 singular values are found to be useful for 2D inversion while typically fewer than 10 are found for 1D. Once the SVD of the combined kernel is obtained, the 2D inversion problem (Eq. (42)) is converted to 1D (Eq. (36)) and the existing algorithms, e.g. Ref. [88] can be used directly. The detailed mathematics are presented in Ref. [61,89].

5.3. Manual compression

There are other techniques to reduce the data size of a 1D, 2D or 3D experiment without using SVD. For example, one may choose to use only a fraction of the data points to perform the inversion in order to reduce the data size. However, this simple method will miss a lot of signal. The so-called window-sum technique manually compresses the CPMG data before optimization and it has been used for Laplace inversion of up to 4D data [90–92].

The basic idea of the window-sum technique is to realize that for a multi-exponential decay signal, the signal at time t is contributed mostly by those signals with decay time constant close or longer than time t . The signal of much shorter decay time constant has essentially decayed away. This means that the signal does not have a rapid time variation and the only potential time scale is t or longer. Thus, a CPMG echo train data can be lumped into a series of bins, and the data of each bin can be a sum (or a mean) of the original echoes. As the decay time t increases, the size of the bin may increase too.

$$Sc_i = \sum_j^{n_i} S_j \quad (43)$$

where S_j is the j th (original) echo data, n_i is the size of the i th bin, and Sc_i is the compressed i th data. The corresponding compressed kernel Kc is

$$Kc_{ij} = \sum_k^{n_i} K_{kj} \quad (44)$$

where K_{kj} was the original kernel. For example, for 1D inversion of T_2 , $K_{kj} = \exp(-t_k/T_{2j})$. Now the compressed equation is

$$Sc = Kc \cdot F \quad (45)$$

and F is the same as in Eq. (36).

5.4. Maximum entropy method (MEM)

In addition to the method outlined in Section 5.2, maximum entropy method has also been used for the MD Laplace inversion [92,93]. The principle of maximum entropy (maxent) is that out of all the solutions that satisfy the data equation, e.g. Eq. (36), the most likely solution is the one that maximizes the entropy function, $S = -\int \mathcal{F}(T) \ln[\mathcal{F}(T)]dT$, and $\mathcal{F}(T) > 0$. Thus the result is

$$F_{ME}(T) = (1/Z) \exp\left(\sum_n \lambda_n S_n V_n\right), \quad (46)$$

where Z is a normalization constant, λ_n is the set of coefficients corresponding to the solution $F \approx \sum_n \lambda_n V_n$. Strictly speaking, Eq. (46) is only true when the λ 's are the solution, however, it can be used as a recursive relationship to obtain the solution iteratively. Even though the formal MEM solution appears to have avoided the regularization parameter choice (e.g. FLI algorithm, Section 5.2), it however requires a predetermined number of singular values to be included in the summation, Eq. (46), based on the SNR. It can be

shown that the regularization corresponds to a smooth cutoff of the singular values, in contrast to the abrupt cutoff.

6. Challenges and new directions

The field of MRPM is ultimately driven by its applications. However, the history has indicated that it is insufficient to simply borrow methods and technologies developed for other NMR applications. The material complexity as well as the unique deployment requirement deserve dedicated R&D for this field. For example, the development of commercial well-logging technology is the result of many years of academic and commercial efforts and resources in theory, instrumentation, and interpretation. I will discuss a few directions of future research in three broad areas.

6.1. Theory and methods

Comparing to the complexity of porous media, our current understanding of the pore space is still very limited. Using the language of eigenmodes, NMR is essentially detecting the ground state alone. For example, the relaxation experiments typically measure the decay time constant of the ground state which is determined by average SVR and not sensitive to pore shape. In a porous rock, the pore space is typically characterized by a pore body and pore throat which connects the pore bodies. The permeability of a rock is highly dependent on pore throat size, which can be very different from the average pore SVR. No current NMR methods are capable to independently determine permeability without laboratory calibration using core samples.

For the related field of medical MRI, it is important to characterize tissue properties both in static structure and dynamics such as blood flow for pathological diagnostics and disease/treatment monitoring. There has been extensive development in this area such as DTI and DSI that have revolutionized our understanding of brain structure, in particular the connectivity of tissues. Such techniques should be applied to study inorganic porous media.

Another area of recent growth is numerical simulation of porous media based on high-resolution X-ray tomography [94]. Rapid improvement of X-ray equipment allows routine acquisition of 3D tomography of porous materials down to 1–2 μm resolutions for a sample of a few millimeter in size. However, geological samples are often heterogeneous and larger samples are likely to contain structures not present in smaller one. It is crucial that such numerical technique can be scaled up to a few cc volume in order to have a meaningful comparison with NMR data.

It is the opinion of this author that the following directions deserve more attention:

- **Novel pulse sequence to detect pore structure**
Pore orientation and anisotropy are important aspects of pore space. DTI and DSI techniques could be applied to map the anisotropy of pore space in order to better understand tortuosity and permeability. Similarly, isotropy of diffusion coefficient [95] could allow better characterization of pore sizes and distribution. A new class of experiments that uses two pairs of PFG pulses (double-PFG) [96–98] has shown interesting features to characterize isotropic and anisotropic pore space. Combining propagator and relaxation [99,100] is another interesting approach. While diffusion is limited in the probing length scale, flow will be able to yield distance larger than grain sizes. Coupling flow to NMR propagator and other methods could be very interesting to probe pore structure over a much large scale.
- **Diffusion dynamics in real pore systems**
We have argued that it is highly desirable to move beyond the ground state and the concept of average SVR in order to

describe more details of the pore space. Combining the advance in high-resolution X-ray tomography and high computing power, it may be feasible now to perform numerical calculation of the eigenmodes and understand the diffusion dynamics in the 2D experiments such as T_1 – T_2 , and T_2 – T_2 , etc. These calculations could become the guide for more sophisticated experiment as well as developing an interpretation of NMR results. Even though the basic theory of diffusion dynamics can be formulated and evaluated, interpretation of experimental results from real materials remains challenging.

- Internal magnetic field and its application

Compared to relaxation and diffusion, internal magnetic field is a relatively new piece of physics that is closely related to the pore structure. Because it exhibits a strong correlation at the pore size scale, it is unique and could be used to obtain pore sizes. In addition, it can be used to excite the higher eigenmodes [101].

- Inversion

Obtaining a spectrum of relaxation or diffusion is equally important as inversion Fourier transform for spectroscopy. However, the mathematics of Laplace inversion is more complex and requires significantly more computing resources. Also due to the nonlinear nature of the inversion, the conventional algorithms (based on the concept of regularization) only derive one plausible solution and do not provide a measure of error and uncertainty.

Statistics based methods (e.g. Monte Carlo [82] or Bayesian approaches [102]) obtain the statistics of the solution space instead of picking a preferred spectrum as the final solution. Such statistics are only constrained by the data with no ad hoc assumption of the properties of the solution (e.g. smoothness). This type of approaches could become the new method of choice for future Laplace inversion analysis.

6.2. Instrumentation

The use of NMR for geological materials often requires a field-worthy portable instrument. Robust and simple operation can be critical for such exploration. In the past two decades, a variety of designs and equipments have been demonstrated [1]. For example, well-logging tools require the measurement volume to be significantly away from the instrument in order to isolate the unaffected formation. This distance is often call DOI, distance or depth of investigation. For an imaging device (such as NMR MOUSE), good spatial resolution and high SNR are important requiring the use of high field permanent magnet and short DOI. Due to the diverse nature of the applications, the requirement and design can be drastically different. This is a significant deviation from the conventional NMR/MRI devices that are of only a few common designs. For a well-logging device, larger DOI would be very desirable; on the other hand for measurement of thin samples, such as surface of paintings, skin, small DOI and high spatial resolution are the key.

Some of the common challenges of MRPM hardware are the following:

- Magnet and design

High magnetic field is often the solution to improve SNR of MR. For inside-out NMR equipments employing permanent magnets, the field is limited to a few hundreds Gauss except for those with very small DOI. Magnetic materials with larger magnetization would be highly desirable, such as superconducting magnets. The practical use of superconductors would be a serious challenge for field application and even more so for the high temperature downhole environment. However, if there is a breakthrough in miniaturized cryogenic coolers, it is conceivable to include superconductors into the magnet design.

- RF design

RF efficiency of an inside-out NMR device is always poor due to the extremely low filling factor. For example, a typical device irradiates a large volume of space where the signal is only generated from a rather localized shell due to the presence of strong magnetic field gradient. Also the B_1 is much weaker at the sample volume than at the surface of the RF probe. It is highly desirable to focus the B_1 field to the sample volume only. Such coil is likely to be much more complex than the simple designs (solenoid, surface loops, etc.) currently in use.

- SNR and sensitivity

Inside-out NMR devices typically exhibit a strong field gradient and thus the detection volume is a thin shell, significantly limiting the SNR. Interestingly, a probe with lower Q may obtain better SNR by enlarging the excitation and detection bandwidth and thus sample volume. This is a scenario quite different from the conventional NMR in uniform field. As a result, a fully broadband system (electronics and probe) may be seriously advantageous. This then requires fully broadband electronics (transmitter and receiver) as well as novel pulses for broadband excitation.

- Surface NMR

Surface NMR (SNMR, or sometime called Magnetic Resonance Sounding, MRS), developed from 1980s [103] is a method to use the Earth magnetic field as the static field and deploys a large coil (up to 100 m) at the surface to detect the signal from the near-surface water body ([104]). Compared to the sophistication of the conventional NMR/MRI, SNMR primarily uses single-pulse FID. Two-pulse sequences have been recently used to measure true T_2 [105] or T_1 [106]. Development of further pulse sequences and the associated hardware will undoubtedly be critical.

- Laboratory NMR equipment

Simulating in situ conditions of porous media could be very challenging in the lab. For example, a typical NMR system for rock core measurement employs a core holder in order to apply pressure and temperature to the sample while RF and gradient coils are placed outside of the core holder [107]. Such a design suffers a low filling factor when the required temperature and pressure are high due to the space needed for a thick wall core holder. Because only non-metallic materials can be used for core holder in such design, the pressure specification is limited. Recent design of putting a RF coil inside the metallic pressure housing is an important innovation to achieve significant higher pressure [108,109]. Compare to pressure, temperature is often a more sensitive parameter determining the spin relaxation and diffusion properties. However, most of the measurements reported in the literature have not explored the temperature dependence. This may be largely due to the lack of suitable commercial sample holders that are designed for alleviated temperatures.

6.3. Applications

The usefulness of MRPM is the ability to measure or infer non-MR knowledge of the porous sample using MR methods. Thus it is crucial to develop the connection between MR parameters with material properties such as porosity, permeability, composition, and fluid–solid interactions, etc. Often such connection is built via a combination of NMR development (e.g. pulse sequence design) and basic physical/chemical research.

- Pore shape and connectivity

Conventional relaxation and diffusion methods are able to characterize the average SVR. Further details of the pore space can be important to determine permeability and flow properties,

they could also provide information on the geological processes for the formation/alteration of the pores (sedimentation and diagenesis). It could also impact the interpretation of other measurements, such as electrical conductivity.

- Solid matrix

Most of the investigation of porous media is focused on the fluids inside the pore space. The mineral phase in fact is just as important in understanding geological processes, mechanical properties, and fluid–solid interaction. However, it may be a challenge to apply solid state NMR methods to such materials. In addition to mineral composition, the physical shape and size of the grains and their contact are very important parameters.

- Wetting, fluid–solid interface, surface relaxivity

One key aspect of the physics governing the overall properties of the fluid-filled porous materials is the fluid–solid interaction at the solid surface. For example, the wetting condition (e.g. oil or water wet) has a large impact on the amount of residue oil left in a oil reservoir after initial production. It could be sensitively dependent on the properties of both the fluid and solid phases. NMR surface relaxivity in fact is a sensitive tool to study such interaction at the molecular scale.

- Temperature dependence

Many of the relaxation phenomena are temperature sensitive due to the molecular interactions between the fluid molecules and with the surfaces. For example, bulk relaxation and surface relaxivity can be temperature dependent. So is the diffusion coefficient of fluids. Such temperature dependence is critical for interpretation of NMR results for fluid or PM properties. Furthermore, the overall diffusion dynamics (Section 4) depend on all such factors (diffusion coefficient, surface relaxivity, bulk relaxation). The literature on temperature dependence of MRPM is in fact very limited [110,63].

- Flow

The non-invasive methods of NMR and MRI to probe inside an opaque sample with molecular sensitivity is unrivaled in the laboratory and in the field. This is particularly true for the study of fluid flow in porous media where majority of the literature on such topics only monitors the effluent discharge. For example, models for filtration can be tested directly by MRI of the trapping phase. This can be particularly interesting for studying transport of nano-size contaminants. Spatial configuration and network of the two fluid phases is dependent on fluid–fluid, fluid–surface interactions and flow history. Understanding of such multi-phase system is very important to understand and characterize oil production (such as EOR, enhanced oil recovery).

Flow is also an important topic for biological systems such as blood flow, perfusion, oxygen and nutrient transport, tumor growth, and drug delivery. Ideas and methods developed in either field could become inspiration for others. The complex hierarchical structure of the blood vessel network will likely prove to be as challenging as the multiphase flow phenomenon in inorganic porous media.

- Low-cost logging tools for prospecting water resource

MR well-logging is a well-established industry with significant commercial support for research, engineering and deployment. Even though the same logging tools can be used in water wells, the high cost of the instrument and their deployment makes it inaccessible for water management. Compared to the deep wells at temperatures up to 200 °C and pressure 2000 atm, the operating environment of a water well is benign and the wells are much shallower (<200 m). A simpler MR logging tool (e.g. Ref. [111,112]) and the associated inexpensive deployment strategy could significantly improve water management.

7. Conclusion

MRPM touches many scientific disciplines both in its technical development and applications. It has its roots in the molecular interactions and dynamics and applications as diverse as modern life itself, and it is built from the foundation of MR science, technology, and methodology. In recent years, it has become clear that simply borrowing from existing MR literature is insufficient and dedicated development for MRPM is required in order to address the profound and distinct needs of the field. This development will inevitably include both new MR methodology as well as understanding the basic physics and chemistry of the materials.

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